Stochastic Processes

Homework 1: Complete Step-by-Step Solutions

Lecture 1 | 20 problems | 100 points

Reading alignment

Sheldon M. Ross, *Introduction to Probability Models*, 10th ed.

- **Primary:** Chapter 1, §§1.1–1.6: sample spaces, counting, probability rules, conditioning, Bayes' theorem, independence.
- **Supporting:** Chapter 2, §§2.1–2.5 and 2.8: random variables, distributions, moments, joint laws, limit theorems.

Instructor note. These solutions follow the distributed Homework 1 problem statements in their original order. Intermediate steps are intentionally explicit. Decimal values are rounded to four places when appropriate. The law of total variance in Problem 20 is a challenge extension.

I. Probability Foundations — 12 points

Problem 1. Sample spaces and events (3 points)

(a) Since components are distinguishable, record the first and second outcomes in order:

$$S = \{WW, WF, FW, FF\}.$$

(b) Exactly one failure occurs in WF or FW, while the first component works in WW or WF:

$$A = \{WF, FW\}, \qquad B = \{WW, WF\}.$$

(c) The union includes outcomes in either event; the intersection includes outcomes in both; the complement is taken relative to S:

$$\boxed{A \cup B = \{WW, WF, FW\}}, \quad \boxed{A \cap B = \{WF\}}, \quad \boxed{A^c = \{WW, FF\}}.$$

Problem 2. Axioms and complements (3 points)

(a) Inclusion–exclusion gives

$$P(A \cup B) = 0.62 + 0.47 - 0.25 = \boxed{0.84}.$$

By De Morgan's law, $A^c \cap B^c = (A \cup B)^c$, hence

$$P(A^c \cap B^c) = 1 - 0.84 = \boxed{0.16}.$$

(b) Partition A into $A \cap B$ and $A \cap B^c$:

$$P(A \cap B^c) = P(A) - P(A \cap B) = 0.62 - 0.25 = \boxed{0.37}.$$

(c) Mutually exclusive events have $P(A \cap B) = 0$, whereas here $P(A \cap B) = 0.25 > 0$. Thus **they are not mutually exclusive**.

Problem 3. Counting outcomes (3 points)

(a) Each die has six possibilities, and order distinguishes the dice: $6^3 = \boxed{216}$.

(b) Each die avoids a six with probability 5/6. Independence gives

$$P(\text{no sixes}) = \left(\frac{5}{6}\right)^3 = \boxed{\frac{125}{216} \approx 0.5787}.$$

(c) The event "at least one six" is the complement:

$$P(\text{at least one six}) = 1 - \frac{125}{216} = \boxed{\frac{91}{216} \approx 0.4213}.$$

Problem 4. Ordered versus unordered sampling (3 points)

(a) For ordered choices without replacement, there are 8 choices, then 7, then 6:

$$P(8, 3) = 8 \cdot 7 \cdot 6 = \boxed{336}.$$

(b) Unordered groups number $\binom{8}{3} = 8!/(3!5!) = \boxed{56}$.

(c) To contain both defective sensors, choose those two and one of the remaining six good sensors. There are $\binom{2}{2}\binom{6}{1} = 6$ favorable groups. Thus

$$P(\text{both defective}) = \frac{6}{56} = \boxed{\frac{3}{28} \approx 0.1071}.$$

II. Conditional Probability and Independence 16 points

Problem 5. Sequential sampling (4 points)

(a) There are 12 balls initially. After a red draw, 5 red remain among 11:

$$P(R_1 \cap R_2) = \frac{6}{12}\frac{5}{11} = \boxed{\frac{5}{22} \approx 0.2273}.$$

(b) Given the first is red, all four blue balls remain:

$$P(B_2 \mid R_1) = \boxed{\frac{4}{11} \approx 0.3636}.$$

(c) The orders RB and BR are disjoint:

$$\begin{aligned} P(\text{one red, one blue}) &= P(RB) + P(BR) \\ &= \frac{6}{12}\frac{4}{11} + \frac{4}{12}\frac{6}{11} \\ &= \frac{48}{132} = \boxed{\frac{4}{11} \approx 0.3636}. \end{aligned}$$

Problem 6. Total probability (4 points)

(a) Let D denote a defect and L_A, L_B, L_C denote the production lines. These line events form a partition, so

$$D = (D \cap L_A) \cup (D \cap L_B) \cup (D \cap L_C),$$

a disjoint union.

(b) Apply total probability, weighting each line's defect rate by its production share:

$$\begin{aligned} P(D) &= P(D \mid L_A)P(L_A) + P(D \mid L_B)P(L_B) + P(D \mid L_C)P(L_C) \\ &= 0.01(0.50) + 0.03(0.30) + 0.05(0.20) \\ &= 0.005 + 0.009 + 0.010 = \boxed{0.024}. \end{aligned}$$

The overall defect rate is 2.4%.

(c) The unweighted average $(0.01 + 0.03 + 0.05)/3 = 0.03$ assumes each line produces one third of devices. Actual production shares differ, so it is not the correct probability for a random device.

Problem 7. Bayesian inspection (4 points)

(a) Let F mean "flagged." From Problem 6, $P(D) = 0.024$ and $P(D^c) = 0.976$. Therefore

$$\begin{aligned} P(F) &= P(F \mid D)P(D) + P(F \mid D^c)P(D^c) \\ &= 0.96(0.024) + 0.04(0.976) \\ &= 0.02304 + 0.03904 = \boxed{0.06208 \approx 0.0621}. \end{aligned}$$

(b) Bayes' theorem reverses the conditioning:

$$P(D \mid F) = \frac{P(F \mid D)P(D)}{P(F)} = \frac{0.02304}{0.06208} = \boxed{\frac{72}{194} \approx 0.3711}.$$

(c) Only 2.4% of devices are defective, so the large population of nondefective devices generates many false flags. A flagged device has about a 37.1% chance of being defective, despite the 96% sensitivity.

Problem 8. Independence versus exclusion (4 points)

(a) Independence requires $P(A \cap B) = P(A)P(B)$. Here $0.4 \cdot 0.5 = 0.2 = P(A \cap B)$, so A **and** B **are independent**.

(b) Mutual exclusion requires $P(A \cap B) = 0$; here it equals 0.2, so **they are not mutually exclusive**.

(c) Suppose A, B are mutually exclusive and $P(A), P(B) > 0$. Then $P(A \cap B) = 0$ but $P(A)P(B) > 0$. Hence $P(A \cap B) \neq P(A)P(B)$, contradicting independence. Therefore two positive-probability mutually exclusive events cannot be independent. □

III. Random Variables and Distributions 20 points

Problem 9. A discrete probability law (5 points)

(a) Normalization requires

$$1 = \sum_{k=0}^{3} c(k+1) = c(1+2+3+4) = 10c, \qquad \boxed{c = \frac{1}{10}}.$$

Thus the four probabilities are $0.1, 0.2, 0.3, 0.4$.

(b) Add the relevant PMF entries:

$$P(X \geq 2) = 0.3 + 0.4 = \boxed{0.7}, \qquad P(X < 2) = 0.1 + 0.2 = \boxed{0.3}.$$

(c) The CDF accumulates mass at each support point:

$$\boxed{F_X(x) = \begin{cases} 0, & x < 0, \\ 0.1, & 0 \leq x < 1, \\ 0.3, & 1 \leq x < 2, \\ 0.6, & 2 \leq x < 3, \\ 1, & x \geq 3. \end{cases}}$$

Problem 10. Binomial reliability (5 points)

(a) Each independent component is a Bernoulli trial with success probability 0.9. Therefore

$$\boxed{X \sim \text{Binomial}(5, 0.9)}.$$

(b) The event $X \geq 4$ consists of $X = 4$ and $X = 5$:

$$\begin{aligned} P(X \geq 4) &= \binom{5}{4}(0.9)^4(0.1) + \binom{5}{5}(0.9)^5 \\ &= 5(0.6561)(0.1) + 0.59049 \\ &= 0.32805 + 0.59049 = \boxed{0.91854 \approx 0.9185}. \end{aligned}$$

(c) Binomial moments give $\boxed{E[X] = 5(0.9) = 4.5}$ and $\boxed{\text{Var}(X) = 5(0.9)(0.1) = 0.45}$.

Problem 11. Poisson arrivals (5 points)

(a) The half-hour interval is 0.5 hours. Its expected count is $8(0.5) = 4$, so $\boxed{N \sim \text{Poisson}(4)}$.

(b) Apply the Poisson PMF and the complement rule:

$$P(N = 3) = e^{-4}\frac{4^3}{3!} = \boxed{\frac{32}{3}e^{-4} \approx 0.1954},$$

$$P(N \geq 1) = 1 - P(N = 0) = \boxed{1 - e^{-4} \approx 0.9817}.$$

(c) A Poisson(4) variable has mean and variance 4. Hence $\boxed{E[N] = 4}$ and $\boxed{\mathrm{SD}(N) = \sqrt{4} = 2}$.

Problem 12. Exponential waiting times (5 points)

(a) With rate 8 per hour, $T \sim \mathrm{Exp}(8)$ in hours:

$$\boxed{f_T(t) = 8e^{-8t} \ (t \geq 0)}, \qquad \boxed{P(T > t) = e^{-8t} \ (t \geq 0)}.$$

The density is zero for $t < 0$; survival equals 1 for $t < 0$.

(b) Fifteen minutes is 1/4 hour:

$$P(T > 15 \text{ min}) = e^{-8(1/4)} = \boxed{e^{-2} \approx 0.1353}.$$

(c) Five minutes is 1/12 hour. Subtract survival probabilities:

$$\begin{aligned} P(5 < T \leq 15 \text{ min}) &= P(T > 1/12) - P(T > 1/4) \\ &= \boxed{e^{-2/3} - e^{-2} \approx 0.3781}. \end{aligned}$$

(d) The mean is 1/8 hour, or $\boxed{E[T] = 60/8 = 7.5 \text{ minutes}}$.

IV. Expectation, Covariance, and Joint Distributions 24 points

Problem 13. Expectation and variance from first principles (6 points)

(a) Weight values by their probabilities:

$$E[X] = (-1)\frac{1}{4} + 0\frac{1}{2} + 2\frac{1}{4} = \boxed{\frac{1}{4}},$$

$$E[X^2] = (-1)^2\frac{1}{4} + 0^2\frac{1}{2} + 2^2\frac{1}{4} = \boxed{\frac{5}{4}}.$$

(b) Apply the second-moment formula:

$$\mathrm{Var}(X) = \frac{5}{4} - \left(\frac{1}{4}\right)^2 = \frac{20-1}{16} = \boxed{\frac{19}{16} = 1.1875}.$$

(c) For $Y = 3X - 2$, use linearity and the variance scaling rule:

$$E[Y] = 3\left(\frac{1}{4}\right) - 2 = \boxed{-\frac{5}{4}}, \quad \mathrm{Var}(Y) = 3^2\frac{19}{16} = \boxed{\frac{171}{16} = 10.6875}.$$

Problem 14. Dependent counts (6 points)

(a) Expectation is linear without independence:

$$E[X + Y] = 10 + 10 = \boxed{20}.$$

Variance includes covariance:

$$\mathrm{Var}(X + Y) = 10 + 10 + 2(3) = \boxed{26}.$$

(b) Since $\mathrm{Cov}(X, -Y) = -\mathrm{Cov}(X, Y)$,

$$\mathrm{Var}(X - Y) = 10 + 10 - 2(3) = \boxed{14}.$$

(c) If independent, covariance would be zero and the sum variance would be 20. Positive covariance makes the two counts tend to move together, increasing sum variability from 20 to 26.

Problem 15. Joint distribution (6 points)

(a) Sum rows for X and columns for Y:

$$P(X = 0) = 0.30 + 0.20 = \boxed{0.50}, \quad P(X = 1) = 0.10 + 0.40 = \boxed{0.50},$$

$$P(Y = 0) = 0.30 + 0.10 = \boxed{0.40}, \quad P(Y = 1) = 0.20 + 0.40 = \boxed{0.60}.$$

(b) Use conditional probability:

$$P(X = 1 \mid Y = 1) = \frac{P(X = 1, Y = 1)}{P(Y = 1)} = \frac{0.40}{0.60} = \boxed{\frac{2}{3}}.$$

(c) Because $XY = 1$ only when $X = Y = 1$, $E[XY] = 0.40$. Also $E[X] = 0.50$ and $E[Y] = 0.60$. Thus

$$\boxed{E[XY] = 0.40}, \qquad \mathrm{Cov}(X, Y) = 0.40 - (0.50)(0.60) = \boxed{0.10}.$$

(d) If independent, $P(X = 1, Y = 1)$ would equal $P(X = 1)P(Y = 1) = 0.30$, not 0.40. Therefore **X and Y are dependent**.

Problem 16. Continuous densities (6 points)

(a) Normalize the density:

$$\begin{aligned} 1 &= c \int_0^2 t(2 - t)\, dt = c \left[t^2 - \frac{t^3}{3} \right]_0^2 \\ &= c \left(4 - \frac{8}{3} \right) = \frac{4c}{3}, \end{aligned}$$

so $\boxed{c = \frac{3}{4}}$.

(b) For $0 \le t \le 2$, integrate from zero to t:

$$F_T(t) = \frac{3}{4} \int_0^t (2u - u^2)\, du = \frac{3}{4} t^2 - \frac{1}{4} t^3.$$

Therefore

$$\boxed{F_T(t) = \begin{cases} 0, & t < 0, \\ \frac{3}{4} t^2 - \frac{1}{4} t^3, & 0 \le t \le 2, \\ 1, & t > 2. \end{cases}}$$

(c) $F_T(1) = 3/4 - 1/4 = 1/2$, hence $\boxed{P(T > 1) = 1/2}$. For the mean,

$$E[T] = \frac{3}{4}\int_0^2 t^2(2-t)\,dt = \frac{3}{4}\left[\frac{2t^3}{3} - \frac{t^4}{4}\right]_0^2$$
$$= \frac{3}{4}\left(\frac{16}{3} - 4\right) = \boxed{1}.$$

(d) A point has zero area under a continuous density: $P(T = 1) = \int_1^1 f_T(t)\,dt = \boxed{0}$. The height $f_T(1) = 3/4$ is a density, not a point probability.

V. Synthesis and Mathematical Reasoning

28 points

Problem 17. Pairwise versus mutual independence (7 points)

(a) The equally likely sample space is $S = \{HH, HT, TH, TT\}$. The events are

$$A = \{HH, HT\}, \quad B = \{HH, TH\}, \quad C = \{HH, TT\}.$$

Each has probability $1/2$.

(b) Pairwise intersections are $A \cap B = \{HH\}$, $A \cap C = \{HH\}$, and $B \cap C = \{HH\}$. Each has probability $1/4 = (1/2)(1/2)$, proving all three pairs are independent.

(c) The triple intersection is also $\{HH\}$, so

$$\boxed{P(A \cap B \cap C) = \frac{1}{4}}, \qquad \boxed{P(A)P(B)P(C) = \frac{1}{8}}.$$

(d) Since $1/4 \neq 1/8$, the events are **not mutually independent**. Pairwise independence checks pairs only; mutual independence also requires the factorization for the triple (and all other subsets).

Problem 18. A two-stage reliability model (7 points)

(a) Let S_i be the supplier event. Exponential survival conditional on supplier i is $P(T > t \mid S_i) = e^{-\lambda_i t}$. Total probability yields

$$\boxed{P(T > t) = 0.7e^{-t/100} + 0.3e^{-t/50}}, \qquad t \geq 0.$$

This is a mixture of exponential survival functions, not generally one exponential survival function.

(b) At $t = 80$,

$$P(T > 80) = 0.7e^{-0.8} + 0.3e^{-1.6} = \boxed{\approx 0.3751}.$$

(c) Apply Bayes' theorem, using survival as the evidence:

$$P(S_1 \mid T > 80) = \frac{P(T > 80 \mid S_1)P(S_1)}{P(T > 80)}$$
$$= \frac{0.7e^{-0.8}}{0.7e^{-0.8} + 0.3e^{-1.6}} = \boxed{\approx 0.8385}.$$

Surviving a long time makes the lower-failure-rate supplier more likely.

(d) Conditional means are $E[T \mid S_1] = 100$ and $E[T \mid S_2] = 50$ hours. Total expectation gives

$$E[T] = 0.7(100) + 0.3(50) = \boxed{85 \text{ hours}}.$$

Problem 19. Law of large numbers and the CLT (7 points)

(a) By linearity, $E[\overline{X}] = (1/400)\sum_{i=1}^{400} 12 = \boxed{12 \text{ minutes}}$. Independence removes covariance terms:

$$\mathrm{Var}(\overline{X}) = \frac{1}{400^2}\sum_{i=1}^{400} 16 = \frac{16}{400} = \boxed{0.04 \text{ min}^2}.$$

Thus $\mathrm{SD}(\overline{X}) = 0.2$ minutes.

(b) The CLT suggests $\overline{X} \approx N(12, 0.04)$. Standardize both endpoints:

$$z_L = \frac{11.6 - 12}{0.2} = -2, \qquad z_U = \frac{12.4 - 12}{0.2} = 2.$$

Consequently

$$P(11.6 \le \overline{X} \le 12.4) \approx \Phi(2) - \Phi(-2) = \boxed{2\Phi(2) - 1 \approx 0.9545}.$$

This is an approximation, since the service-time distribution is unspecified.

(c) The **law of large numbers** concerns convergence of sample averages toward 12 as sample size increases. The **central limit theorem** describes the approximate shape and scale of the sampling fluctuations, permitting the numerical probability in (b).

Problem 20. Integrated service-center model (7 points)

(a) Two hours at rate 6 per hour yields $\boxed{N \sim \text{Poisson}(12)}$. Therefore $\boxed{E[N] = 12}$ and $\boxed{\mathrm{Var}(N) = 12}$.

(b) Given $N = n$, each arrival independently needs diagnostics with probability $p = 0.2$. Thus

$$\boxed{D \mid (N = n) \sim \text{Binomial}(n, 0.2)}, \quad \boxed{E[D \mid N = n] = 0.2n},$$

and $\mathrm{Var}(D \mid N = n) = n(0.2)(0.8) = 0.16n$.

(c) Total expectation gives

$$E[D] = E[E[D \mid N]] = E[0.2N] = 0.2(12) = \boxed{2.4}.$$

(d) Apply the supplied law of total variance:

$$\begin{aligned}\mathrm{Var}(D) &= E[\mathrm{Var}(D \mid N)] + \mathrm{Var}(E[D \mid N]) \\ &= E[0.16N] + \mathrm{Var}(0.2N) \\ &= 0.16(12) + (0.2)^2(12) \\ &= 1.92 + 0.48 = \boxed{2.4}.\end{aligned}$$

The first term reflects randomness of diagnostic flags for a fixed arrival count; the second reflects randomness in the arrival count.

(e) Conditional on $N = n$, no one needs a diagnostic check with probability $(1 - 0.2)^n = 0.8^n$. Sum over the Poisson count:

$$\begin{aligned} P(D = 0) &= \sum_{n=0}^{\infty} P(D = 0 \mid N = n) P(N = n) \\ &= \sum_{n=0}^{\infty} (0.8)^n e^{-12} \frac{12^n}{n!} \\ &= e^{-12} \sum_{n=0}^{\infty} \frac{(9.6)^n}{n!} \\ &= e^{-12} e^{9.6} = \boxed{e^{-2.4} \approx 0.0907}. \end{aligned}$$

There is about a 9.1% chance that no arriving customer requires a diagnostic check during the two-hour interval, including the possibility of zero arrivals.

End of solution guide. The calculations use the definitions and techniques from Lecture 1, with the total-variance identity supplied in Problem 20 as an extension.